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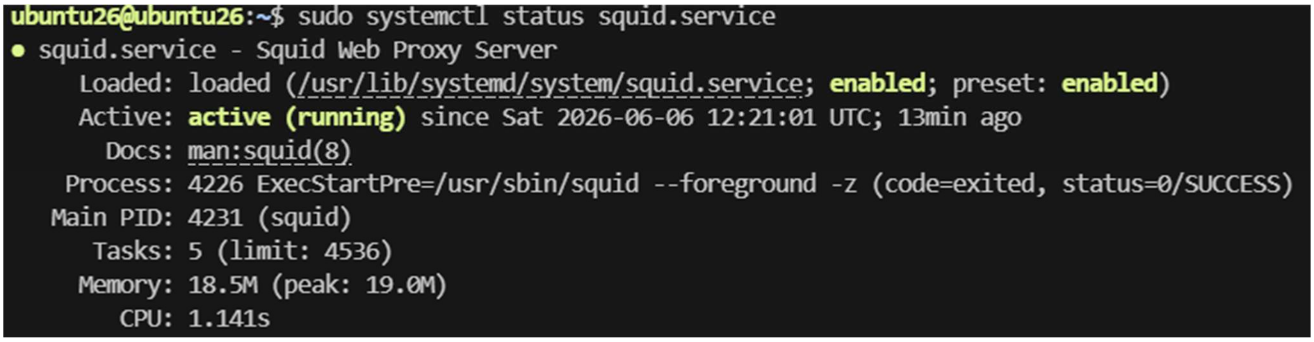
Assignment 10: Proxy server configuration and web filtering

https://github.com/elovric13/eutech/tree/proxy-server-configuration-and-web-filtering-elovric13

1. Install and configure Squid Proxy on a Linux system

Squid was installed on an Ubuntu virtual machine using the apt package manager. After installation, the service was started and confirmed as active. The default configuration file at /etc/squid/squid.conf



```
ubuntu26@ubuntu26:~$ sudo systemctl status squid.service
● squid.service - Squid Web Proxy Server
     Loaded: loaded (/usr/lib/systemd/system/squid.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-06-06 12:21:01 UTC; 13min ago
       Docs: man:squid(8)
    Process: 4226 ExecStartPre=/usr/sbin/squid --foreground -z (code=exited, status=0/SUCCESS)
   Main PID: 4231 (squid)
      Tasks: 5 (limit: 4536)
     Memory: 18.5M (peak: 19.0M)
        CPU: 1.141s
```

Figure 1: systemctl status squid

2. Define ACL (Access Control Lists)

Access Control Lists were defined in the Squid configuration file to control which websites clients can access.

- **Allow access to specific websites only**: An ACL named allowed_sites was created listing permitted domains.

```
acl allowed_sites dstdomain .google.com .wikipedia.org .github.com

http_access allow allowed_sites
```

- **Block social websites**: An ACL named blocked_social was created listing social media

```
acl blocked_social dstdomain .facebook.com .instagram.com .twitter.com
.youtube.com .tiktok.com

http_access deny blocked_social
```

```
# INSERT YOUR OWN RULE(S) HERE TO ALLOW ACCESS FROM YOUR CLIENTS
#
include /etc/squid/conf.d/*.conf

# Allow specific websites
acl allowed_sites dstdomain .google.com .wikipedia.org .github.com

# Block social websites
acl blocked_social dstdomain .facebook.com .instagram.com .twitter.com .youtube.com .tiktok.com

# For example, to allow access from your local networks, you may uncomment the
# following rule (and/or add rules that match your definition of "local"):

http_access deny blocked_social
http_access allow allowed_sites

access_log /var/log/squid/access.log squid

# http_access allow localnet
# And finally deny all other access to this proxy
http_access deny all
```

Figure 2: squid.conf showing the full ACL configuration

- **Record navigation logs**: Squid logs all requests by default. The access log location was explicitly set in the configuration to /var/log/squid/access.log. Each log entry includes the timestamp, response code, request method, destination domain, and whether the request was allowed or denied.

```
access_log /var/log/squid/access.log squid
```

3. **Configure a client browser to connect through the proxy**

The proxy was configured at the system level on the same Ubuntu machine by setting environment variables to route HTTP and HTTPS traffic through Squid on port 3128, which is the default Squid port.

```
export http_proxy="http://127.0.0.1:3128"

export https_proxy="http://127.0.0.1:3128"
```

All curl requests were sent explicitly through the proxy using the --proxy flag to ensure traffic was routed through Squid and subject to the ACL rules.

4. Test and verify that restrictions and logging work as expected

Tests were performed using curl with the --proxy flag to route requests through Squid. Both allowed and blocked sites were tested over HTTP.

URL tested	Expected result	Actual result
https://www.google.com	200 OK — allowed	200 OK
https://www.wikipedia.org	200 OK — allowed	200 OK
https://www.github.com	200 OK — allowed	200 OK
https://www.youtube.com	403 Forbidden — blocked	403 Forbidden
https://www.facebook.com	403 Forbidden — blocked	403 Forbidden
https://www.instagram.com	403 Forbidden — blocked	403 Forbidden
https://www.tiktok.com	403 Forbidden — blocked	403 Forbidden

```
ubuntu26@ubuntu26:~$ curl -I --proxy http://127.0.0.1:3128 https://www.google.com
HTTP/1.1 200 Connection established

HTTP/2 200
```

```
ubuntu26@ubuntu26:~$ curl -I --proxy http://127.0.0.1:3128 https://www.wikipedia.org
HTTP/1.1 200 Connection established

HTTP/2 200
```

```
ubuntu26@ubuntu26:~$ curl -I --proxy http://127.0.0.1:3128 https://www.github.com
HTTP/1.1 200 Connection established
```

Figure 3: curl results showing 200 OK for allowed sites

```
ubuntu26@ubuntu26:~$ curl -I --proxy http://127.0.0.1:3128 https://www.youtube.com
HTTP/1.1 403 Forbidden
Server: squid/6.14
```

```
ubuntu26@ubuntu26:~$ curl -I --proxy http://127.0.0.1:3128 https://www.facebook.com
HTTP/1.1 403 Forbidden
Server: squid/6.14
```

```
ubuntu26@ubuntu26:~$ curl -I --proxy http://127.0.0.1:3128 https://www.instagram.com
HTTP/1.1 403 Forbidden
Server: squid/6.14
```

```
ubuntu26@ubuntu26:~$ curl -I --proxy http://127.0.0.1:3128 https://www.tiktok.com
HTTP/1.1 403 Forbidden
Server: squid/6.14
```

Figure 4: curl results showing 403 Forbidden for blocked sites

After running the curl tests, the Squid access log was checked to confirm all requests were recorded. Allowed requests appeared with TCP_MISS/200 status and blocked requests appeared with TCP_DENIED/403 status.

```
sudo tail -f /var/log/squid/access.log
```

```
1781002296.627    493 127.0.0.1 TCP_TUNNEL/200 5740 CONNECT www.google.com:443 - HIER_DIRECT/142.251.154.119 -
1781002303.201    682 127.0.0.1 TCP_TUNNEL/200 5121 CONNECT www.wikipedia.org:443 - HIER_DIRECT/185.15.58.224 -
1781002309.432    668 127.0.0.1 TCP_TUNNEL/200 3463 CONNECT www.github.com:443 - HIER_DIRECT/140.82.121.4 -
1781002314.026     24 127.0.0.1 TCP_DENIED/403 3434 CONNECT www.facebook.com:443 - HIER_NONE/- text/html
1781002319.347     28 127.0.0.1 TCP_DENIED/403 3436 CONNECT www.instagram.com:443 - HIER_NONE/- text/html
1781002327.936     38 127.0.0.1 TCP_DENIED/403 3432 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1781002335.946   2185 127.0.0.1 TCP_DENIED/403 3430 CONNECT www.tiktok.com:443 - HIER_NONE/- text/html
```

Figure 5: access.log